

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

The website was most likely affected by a TCP SYN Flood, which is a type of Denial-of-Service (DoS) attack. The packet capture shows an unusually large number of TCP SYN requests coming from an unknown IP address. The web server became overwhelmed by these connection requests and was unable to respond to legitimate users, resulting in connection timeout errors.

The evidence indicates a DoS attack rather than a DDoS attack because the malicious traffic appears to originate from a single IP address.

Section 2: Explain how the attack is causing the website to malfunction

A TCP connection normally begins with the three-way handshake:

1. The client sends a SYN packet.
2. The server responds with a SYN-ACK packet.
3. The client completes the connection by sending an ACK packet.

During a SYN Flood attack, the attacker sends a large number of SYN packets to the server without completing the handshake. The server allocates resources for each pending connection while waiting for the final ACK packet. As these half-open connections accumulate, the server's available resources become exhausted.

As a result, legitimate users cannot establish new connections, causing the website to become slow or unavailable and generating connection timeout errors.